

APATITY, Russian Federation

WMO: 222130

Lat: 67.5508N

Lon: 33.3622E

Elev: 134

StdP: 99.73

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.7	-28.9	-34.0	0.2	-31.5	-31.2	0.2	-29.0	10.1	-6.7	9.2	-5.4	0.4	100	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	23.9	16.6	22.0	15.5	19.9	14.5	17.6	22.3	16.2	20.5	15.1	19.0	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	11.1	19.8	14.2	10.3	18.3	13.2	9.6	17.1	50.0	22.7	45.6	20.5	42.7	19.2	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.3	6.4	-34.0	26.5	2.2	2.8	-35.7	28.5	-37.0	30.1	-38.2	31.6	-39.9	33.7		
			-34.5	19.0	2.4	1.8	-36.3	20.3	-37.7	21.3	-39.0	22.3	-40.8	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-14.0	-10.2	-7.1	-1.4	5.3	10.2	14.3	12.1	7.1	0.2	-5.8	-10.5	(6)
(7)		DBStd	10.73	7.83	7.35	5.80	4.50	4.28	4.05	3.63	3.36	3.19	4.46	5.89	7.50	(7)
(8)		HDD10.0	3918	746	566	531	343	156	46	6	17	97	304	473	636	(8)
(9)		HDD18.3	6679	1004	799	789	593	404	245	132	197	338	563	723	894	(9)
(10)		CDD10.0	291	0	0	0	0	10	52	139	80	9	0	0	0	(10)
(11)		CDD18.3	11	0	0	0	0	0	2	7	2	0	0	0	0	(11)
(12)		CDH23.3	78	0	0	0	0	1	11	56	11	0	0	0	0	(12)
(13)		CDH26.7	5	0	0	0	0	0	0	4	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.9	2.8	2.9	3.0	3.2	3.1	2.8	2.5	2.8	2.8	2.7	2.5	(14)
(15)	Precipitation	PrecAvg	760	67	53	53	41	51	66	76	71	66	79	66	70	(15)
(16)		PrecMax	881	103	92	105	94	82	110	120	116	112	148	117	114	(16)
(17)		PrecMin	657	26	24	30	13	25	20	30	28	27	29	29	42	(17)
(18)		PrecStd	61	18	18	17	14	14	24	20	23	21	27	21	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	2.5	4.7	11.7	21.0	24.7	27.2	24.6	16.8	9.5	3.6	1.9	(19)
(20)			MCWB	1.2	1.6	1.7	6.3	12.7	15.6	18.8	18.5	12.9	8.0	2.6	1.1	(20)
(21)		2%	DB	-0.5	1.2	2.4	7.7	17.6	22.3	24.7	21.7	14.7	7.9	2.4	0.4	(21)
(22)			MCWB	-1.4	0.5	0.5	3.8	10.8	14.9	16.8	16.1	11.6	6.5	1.9	-0.2	(22)
(23)		5%	DB	-1.9	0.4	1.2	5.6	15.0	19.7	23.1	19.2	13.2	6.9	1.6	-0.5	(23)
(24)			MCWB	-2.5	-0.1	-0.2	2.8	9.5	13.3	16.3	15.0	10.9	5.8	1.1	-1.1	(24)
(25)		10%	DB	-3.3	-0.8	0.2	3.8	12.3	17.4	21.2	17.4	11.8	5.7	0.7	-1.6	(25)
(26)			MCWB	-3.8	-1.4	-0.8	1.5	7.9	12.1	15.4	14.0	9.9	4.8	0.3	-2.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.7	1.8	6.4	13.4	17.1	19.4	19.5	13.7	8.4	2.9	1.2	(27)
(28)			MCDWB	2.3	2.1	4.0	10.6	19.7	22.5	24.9	24.0	15.5	9.1	3.2	1.8	(28)
(29)		2%	WB	-1.2	0.6	0.8	4.3	11.4	15.5	18.3	16.8	12.1	6.8	2.1	0.0	(29)
(30)			MCDWB	-0.7	1.0	2.1	7.4	16.4	21.0	23.5	20.7	14.1	7.6	2.4	0.3	(30)
(31)		5%	WB	-2.5	-0.1	0.1	2.8	9.9	13.9	16.9	15.4	11.1	6.0	1.2	-1.0	(31)
(32)			MCDWB	-2.0	0.3	1.0	5.0	14.9	18.4	21.1	18.4	12.9	6.9	1.5	-0.6	(32)
(33)		10%	WB	-3.8	-1.4	-0.7	2.0	8.2	12.7	15.7	14.3	10.1	4.8	0.5	-2.0	(33)
(34)			MCDWB	-3.3	-1.0	0.1	3.4	12.1	17.0	20.2	17.0	11.5	5.7	0.8	-1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	6.6	8.1	7.3	7.2	7.4	7.9	6.8	6.0	4.4	5.2	7.4	(35)
(36)			MCDBR	6.8	4.2	7.0	8.1	12.5	11.4	12.7	10.0	8.9	4.7	3.7	5.6	(36)
(37)			MCWBR	6.6	4.1	5.4	5.3	7.1	5.7	6.1	5.4	5.8	3.9	3.5	5.1	(37)
(38)		5% WB	MCDBR	6.6	4.6	5.9	6.5	12.1	9.5	9.9	9.0	7.3	4.3	3.6	5.4	(38)
(39)			MCWBR	6.4	4.5	4.8	4.5	7.1	5.2	5.2	5.1	5.2	3.7	3.6	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.215	0.271	0.309	0.344	0.361	0.359	0.379	0.358	0.326	0.288	N/A	N/A	(40)	
(41)		taud	1.906	2.030	2.068	2.120	2.303	2.420	2.396	2.428	2.482	2.252	N/A	N/A	(41)	
(42)		Ebn at Noon	104	512	708	794	829	844	810	788	726	510	N/A	0	(42)	
(43)		Edh at Noon	12	53	90	114	107	98	98	85	63	43	N/A	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.05	0.49	1.87	3.52	4.55	5.06	4.60	3.19	1.73	0.56	0.10	0.00	(44)	
(45)		RadStd	0.00	0.08	0.18	0.28	0.37	0.48	0.58	0.29	0.15	0.08	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page